

Introduction

Aspirin Usage among adults 50 - 80

1/4 take regularly
(3 or more days a week)

14%

Don't have a history of Cardiovascular Diseases

have a history of Cardiovascular Diseases

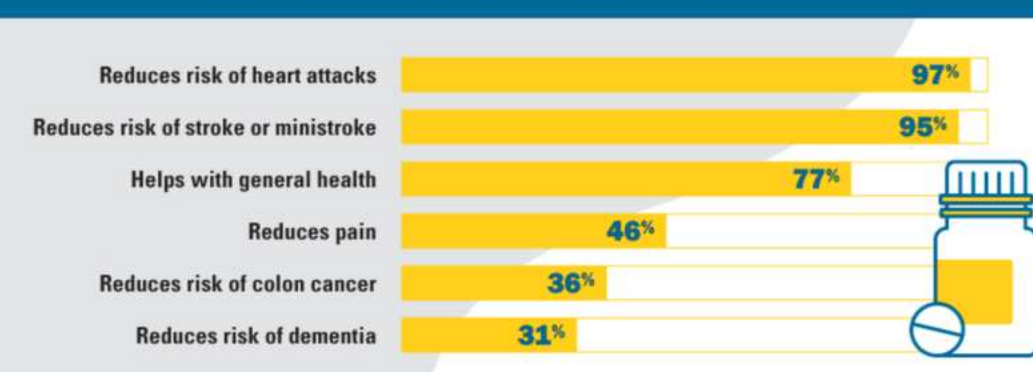
11%

Why is resistance a problem?

- Increases the risk of complications with treatments
- Increase risk within the ER and other medical industries
- Shows a growing pattern of resistance to common medications

NSAIDs, nonsteroidal anti-inflammatory drugs are medications that have been widely believed to be very **resistant** to **tolerance** in patients. In recent years, these medications have been shown to have increased **resistance**. This poses the risk of complications in areas like emergency rooms, where **NSAIDs** like aspirin are used as first-response medication.

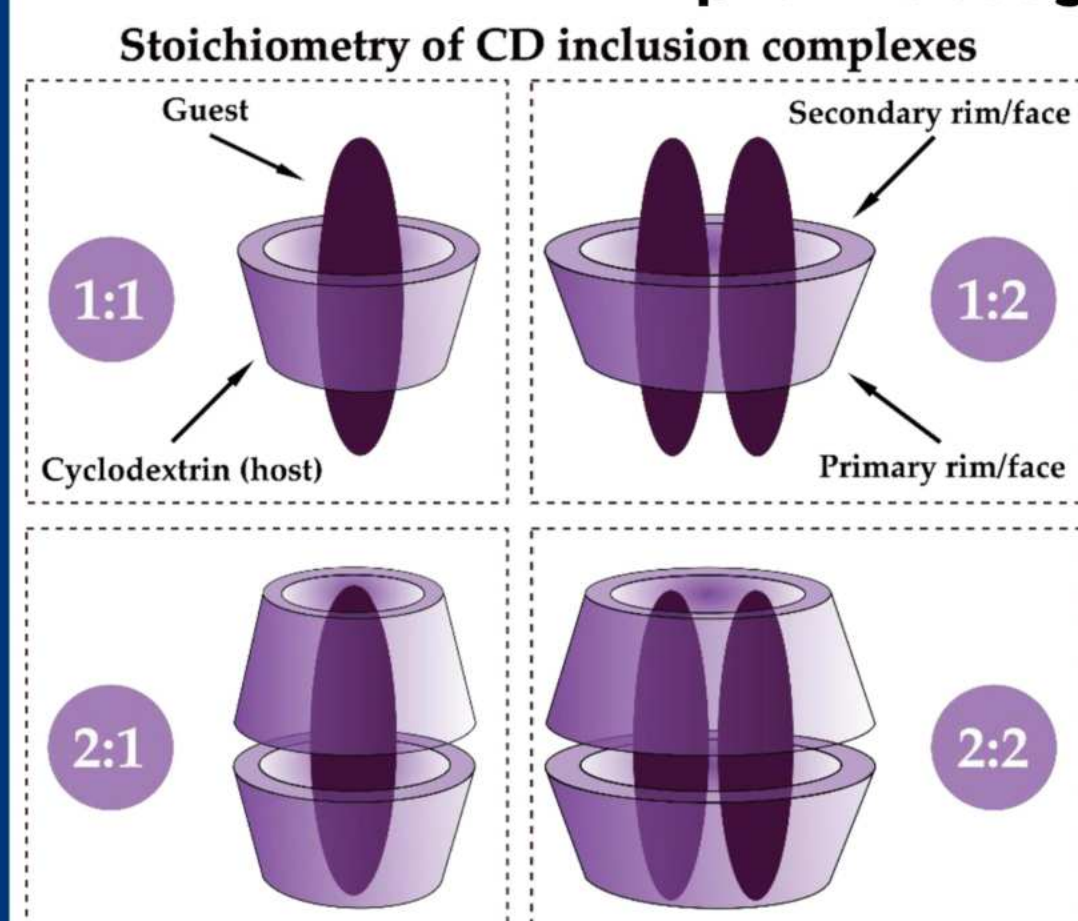
Perceived benefits of aspirin
AMONG ADULTS AGE 50-80 WHO TAKE ASPIRIN REGULARLY



Aspirin Use Among Older Adults. 2024 March 7th. Institute for Healthcare Policy & Innovation. <https://ipih.umich.edu/national-poli-health/aging/national-findings/aspirin-use-among-older-adults>

Solutions to this growing issue are increasing dosage and changing to stronger **NSAIDs**. These methods pose a greater risk due to stronger **NSAIDs**, which come with greater issues, like a lack of bioavailability or harmful side effects.

One possible way to affect **bioavailability** is through cellular **pharmacokinetics**. Changing **pharmacokinetics** can change **pharmacodynamics**.



To make this Change in the cell, **cyclodextrins** can be used in the form of complexes with drugs like aspirin. This creates a cone-shaped molecule with different or similar effects, but due to the shape, intact with barriers differently, like the cell membrane. Causing a possibly greater effect in combating **inflammation**.

Significants

- The primary purpose of the experiment is to find greater alternatives to **NSAIDs** resistant to other than changing dosage and medication due to the risks that can occur.
- Sugammadex** and **SBECD** have been tested with **aspirin** and found an increase in **bioavailability** and passive uptake.
- Aspirin is a common **NSAID** that has seen an increase in resistance due to its common usage in daily life.
- Connects increased uptake to help increase effects within the **inflammation** model using **Lipopolysaccharides** (LPS).

Research Question

If the cyclodextrins, SBECD and Sugammadex, create a complex with aspirin, causing a change in the pharmacokinetics of the aspirin, then the pharmacodynamics of aspirin would change, modeled by the greater decrease in COX-2 protein expression.

The Effects of The Aspirin-Cyclodextrin Complex On The Pharmacodynamics In The Protein Expression of COX-2 In The Cell Line A549

Methodology

Acid Phosphatase Assay

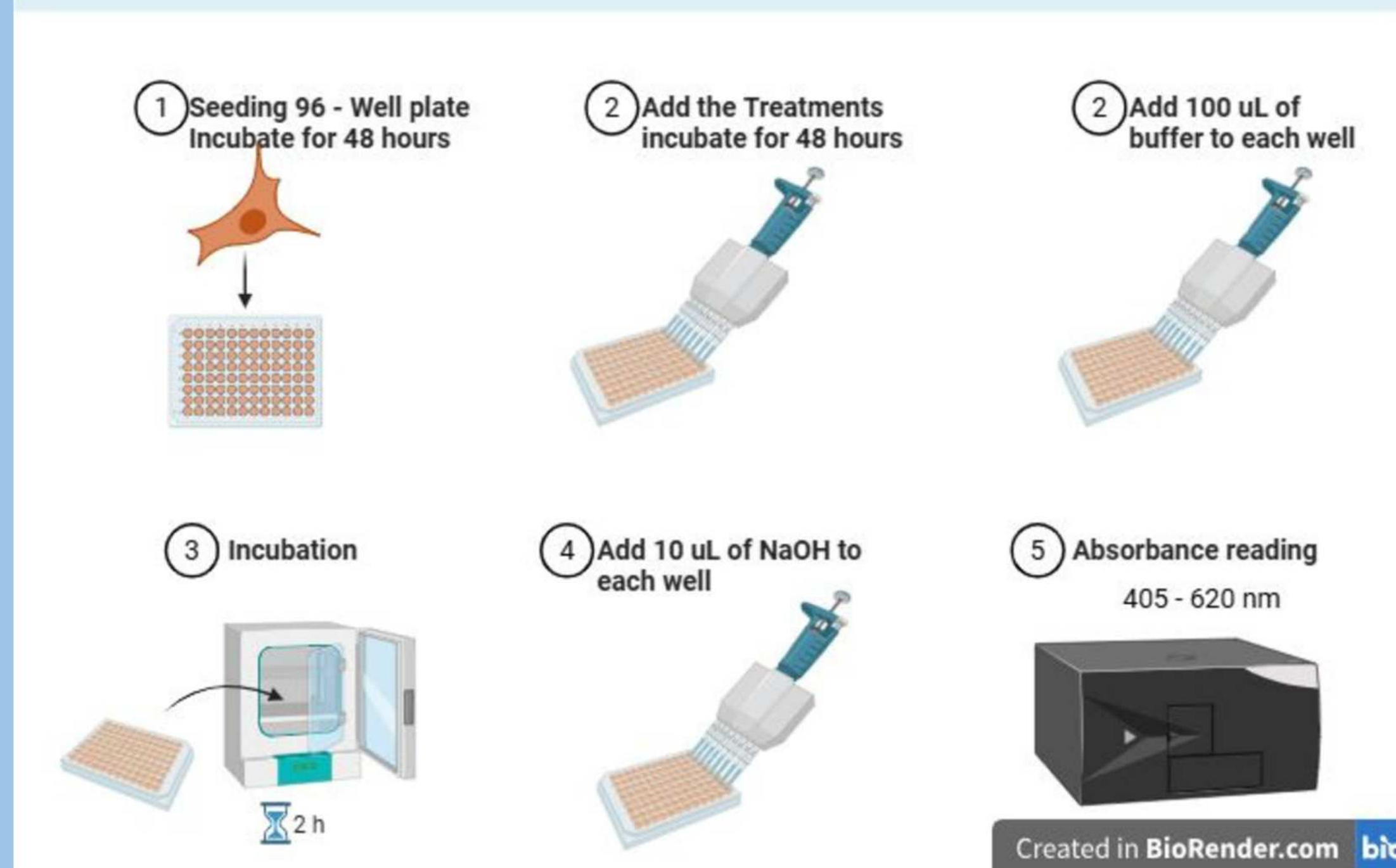


Figure 1 : Being used to measure the cell viability through the expression and reaction of acid phosphatase to evaluate the effects of treatments on cell viability, to correlate with the potential safety and usage of the treatment

IV: Types of Cyclodextrin used in the complex , either SBECD or Sugammadex

DV: Cell viability measured by fluorescence of Acid phosphatase → Acid Phosphatase assay

The expression of COX-2 protein from A549 cells → Western Blot

Western Blotting Experiment

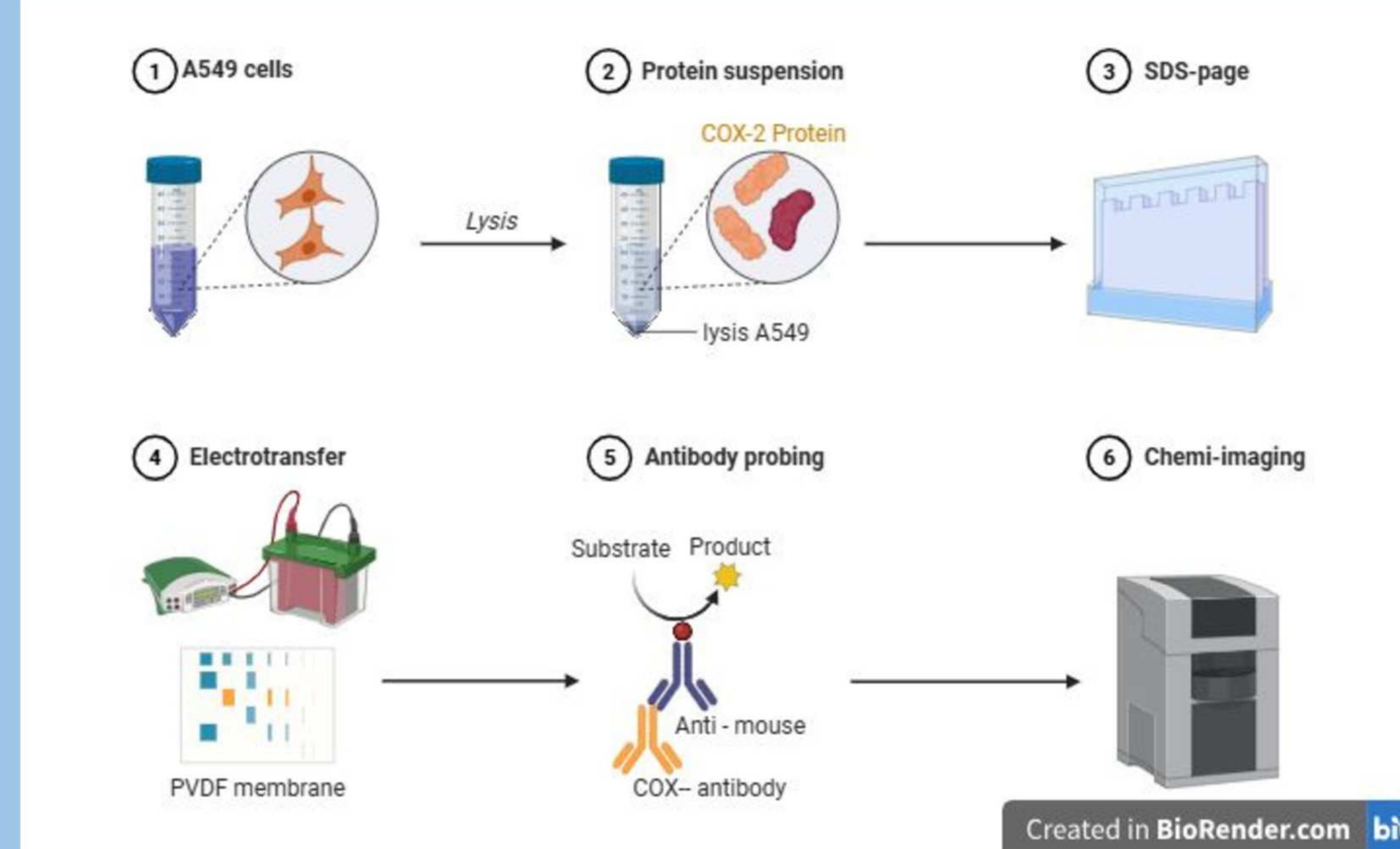


Figure 2 :Being used to measure the expression of the COX-2 protein, imaged, and evaluating the pixel density of the images to see the change with the expression of the treatments

Treatments: 100 uM Aspirin ,100 uM Aspirin-SBECD complex , 100 uM Aspirin - Sugammadex Complex

Controls: Negative = Growth Media, Growth Media and LPS treatment (2ug/ml), vehicle = Water

Constants: A549 Cell line, All are treated with TNF-alpha, Growth media → 10%FBS + 90% F-12 media

Data and Results

Cell Viability

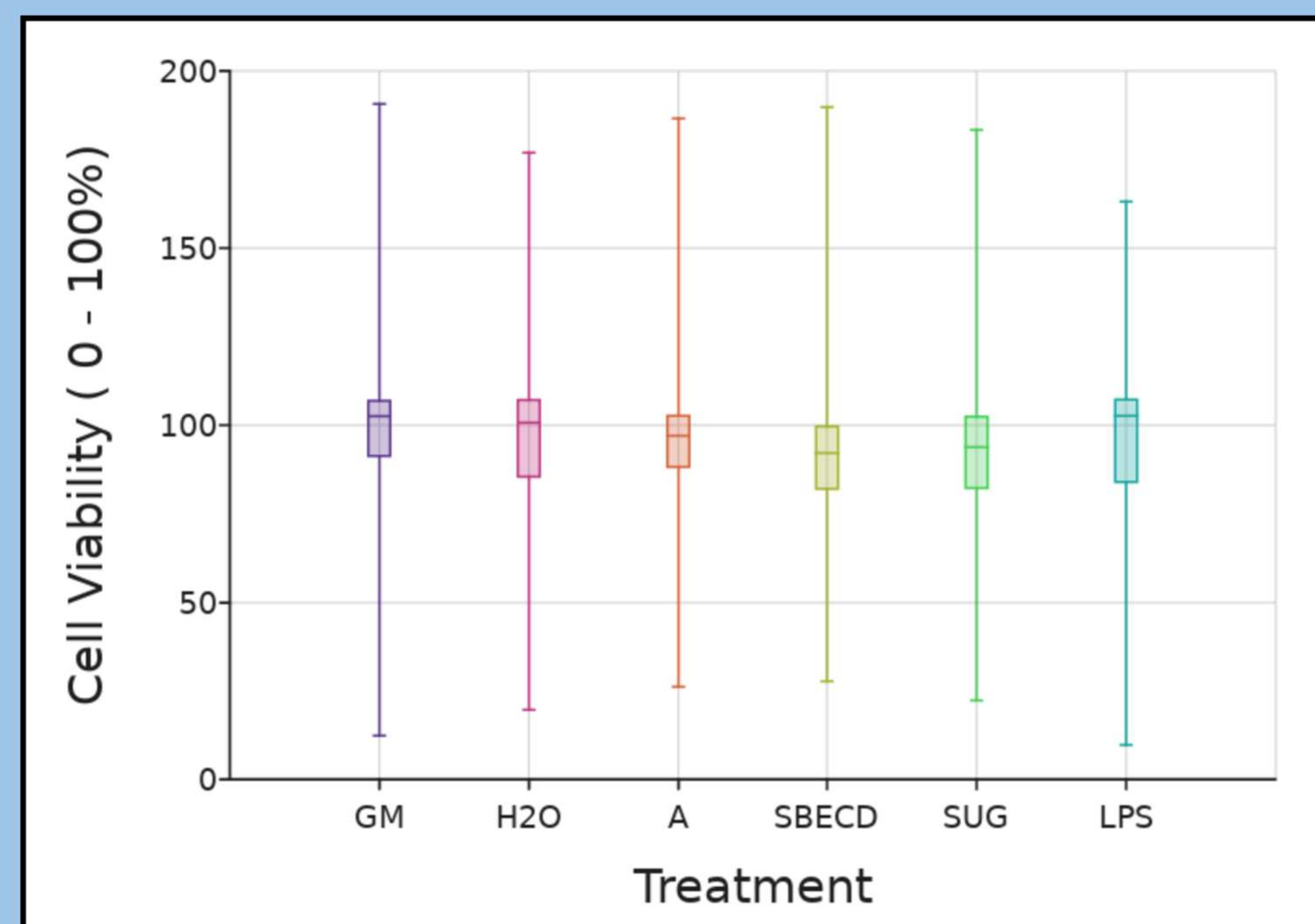


Figure 3 : Graph of compiled Cell viability via Acid phosphatase that looks at the absorbance that is directly correlated to cell availability

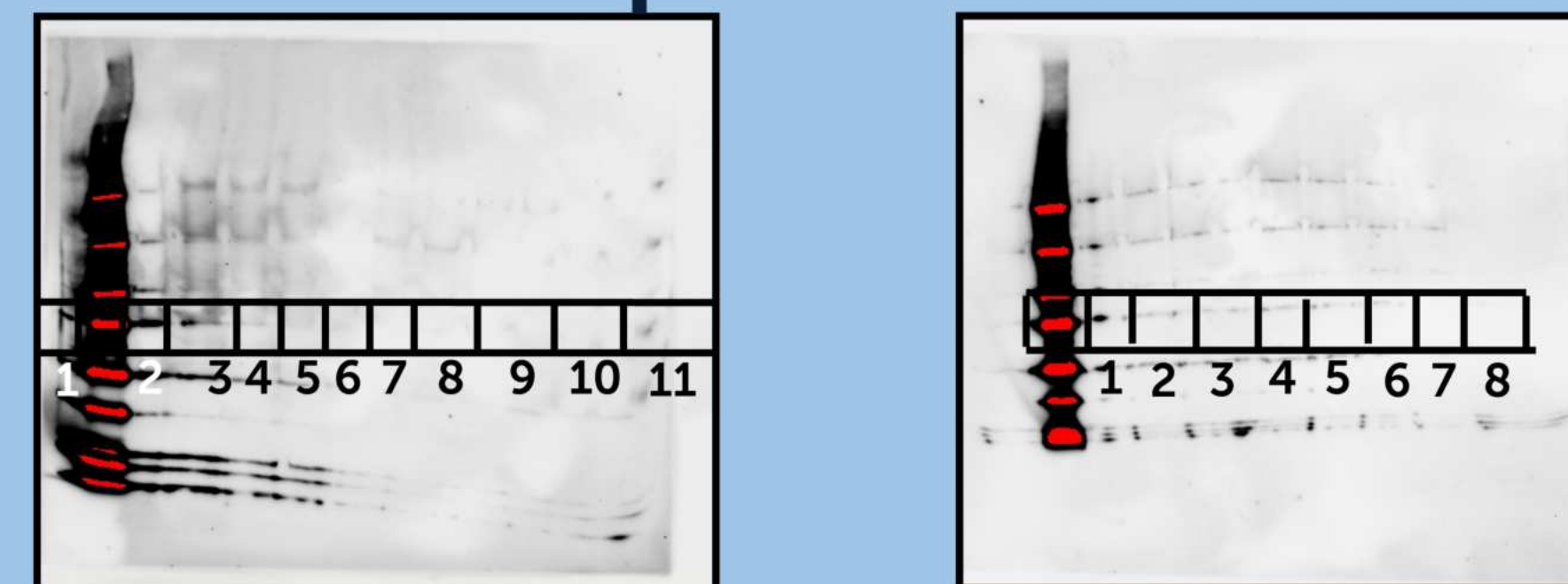
Key: (GM : Growth Media) (H2O : Water) (A : Aspirin) (SBECD: Cyclodextrin Complex SBECD) (SUG : Cyclodextrin Complex Sugammadex) (LPS : Lipopolysaccharides)

Table 1 :

Acid Phosphatase T-Test Results

Post Hoc Comparisons			T-Test
Pairs	P-value	Interpretation	Interpretation of P
H2O - GM		1 Groups are NOT different	A P-value of 0.99
A - GM		1 Groups are NOT different	
SUG - SBECD		1 Groups are NOT different	
SBECD - A		1 Groups are NOT different	
SUG - A		1 Groups are NOT different	

Protien Expressionof COX-2



Figures 4 & 5: Western Blot images used to visualize COX-2 expression and data by the usage of ImageJ for pixel density of the area. Red marks the standard to help find the relative position of the COX-2 protein.

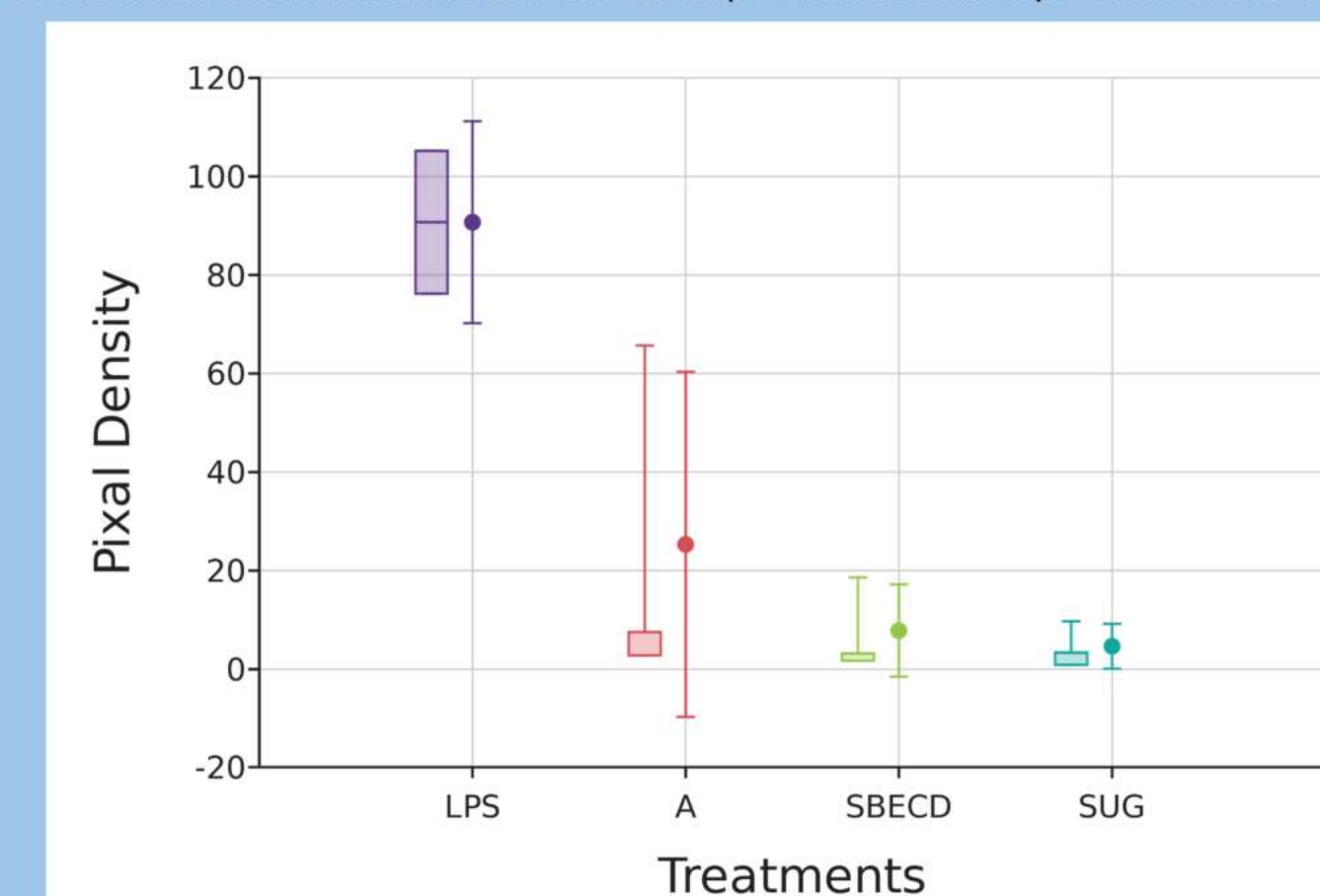


Figure 6 : Graph of compiled Western Blot data showing that the expression of COX-2 is correlated to the pharmacodynamics of Aspirin

Key: (A : Aspirin) (SBECD: Cyclodextrin Complex SBECD) (SUG : Cyclodextrin Complex Sugammadex) (LPS : Lipopolysaccharides)

Table 2 :

Western Blot ANOVA Results

Post Hoc Comparisons			ANOVA
Pairs	P-value	Interpretation	Interpretation of P
A - LPS	0.04	Strong confidence that the groups are	A P-value of 0.01
SBECD - LPS	0.01	Very strong confidence that the	
SUG - LPS	0.01	Very strong confidence that the	
SBECD - A	0.74	No evidence that the groups might be	
SUG - A	0.64	No evidence that the groups might be	
SUG - SBECD	1	NOT different	

Discussion

Further research on the Cyclodextrin Complex on aspirin pharmacokinetics needs to be done.

Acid Phosphatase

Based on the **acid phosphatase** analysis, the cyclodextrin complex of **SBECD** and **sugammadex** does not change cell viability compared to the standalone aspirin and the stimulation of **LPS**. With a P value of 1 on the Hoc pairs and a over all P-value of 0.99 in the T-test indicates that the cell viability was not affected by the addition of the complex or the treatments, likely by chance. Meaning tha that the complexes do not likely impact the safety of the medication, with the addition that the change in protein expression in the Western blot is not due to cell death from the treatment.

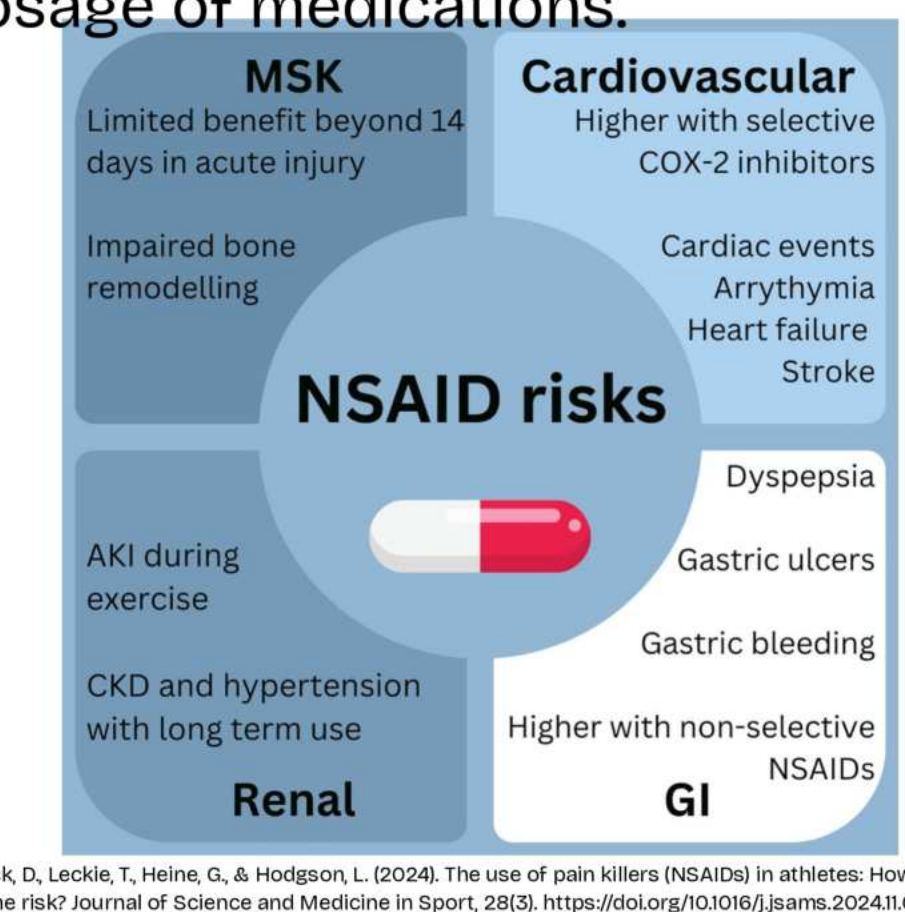
Western Blot

Western blot protein expression provided evidence that the complex between aspirin and the cyclodextrin **SBECD** and **sugammadex** does not change the overall **pharmacodynamics** of aspirin in decreasing inflammation by decreasing the presence of **COX-2**, but does not provide strong evidence for the magnitude. The overall p-value of 0.01 in the ANOVA indicates that there is a change in the **COX-2** expression and that the complexes did not change that aspect, but the p-values of the complexes to the aspirin in the post Hoc test do not provide strong evidence of differences or not.

Future Work

The next largest step to the experiment is using different concentrations of aspirin and its precursors, allowing a more detailed analysis of the impacts of cyclodextrin complex provided to NSAIDs , and possible implications to pharmacological issues like increased drug resistance in patients, and possible positive impacts to pharmacodynamics without changing the dosage of medications.

- Use a more definitive method for complex creation
- Test for passive absorbance and active absorbance
- Test other inflammation models and other COX proteins
- Test different concentrations of different NSAIDs and Cyclodextrins



References

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